

Spotify Music Popularity Analysis

Business Requirement

Business Scenario

Spotify is one of the world's leading music streaming platforms, hosting millions of songs from artists across different genres. Every day, thousands of new songs are added to the platform, making it challenging to understand what type of music attracts listeners and gains popularity.

As a **Junior Data Analyst** at Spotify, your responsibility is to analyze the company's music dataset and identify trends related to artists, genres, albums, release years, and song popularity. The insights generated from your analysis will help the content team understand listener preferences, improve playlist recommendations, and identify successful music categories.

Using Python and Pandas, perform data cleaning, exploratory data analysis (EDA), and visualization to answer key business questions and provide meaningful recommendations.

Project Objective

Analyze the Spotify dataset to:

- Understand the distribution of songs across artists and genres.
- Identify the most popular songs and artists.
- Explore trends based on release year.
- Create meaningful visualizations.
- Generate business insights and recommendations.

Dataset Description

The dataset contains information about **7,000 Spotify songs**.

Dataset Features

Column	Description
Track ID	Unique Song Identifier
Track Name	Name of the Song
Artist(s)	Artist Name
Album	Album Name
Release Date	Song Release Date
Cover Image	Album Cover URL
Popularity	Spotify Popularity Score
Genre	Music Genre

Tasks to Perform

Task 1 – Data Loading

- Import the required libraries.
- Load the dataset using Pandas.
- Display the first five records.
- Display the last five records.
- Find the number of rows and columns.
- Display all column names.
- Check the data types of each column.

Task 2 – Data Exploration

Perform the following analysis:

- Check for missing values.
- Check duplicate records.
- Display summary statistics.
- Find the number of unique artists.
- Find the number of unique albums.
- Find the number of unique genres.

Task 3 – Data Cleaning

Perform the following operations:

- Remove duplicate records (if any).
- Handle missing values.
- Convert **Release Date** into Date format.
- Create a new column named **Release Year**.

Business Questions

Answer the following questions using Python and Pandas.

Artist Analysis

1. Which artist has the highest number of songs?
2. Display the Top 10 artists based on song count.
3. Which artist has the highest average popularity score?

Genre Analysis

4. Which genre has the highest number of songs?
5. Which genre has the highest average popularity?
6. Compare the popularity of different genres.

Album Analysis

7. Which albums contain the highest number of songs?
8. Which albums have the highest average popularity?

Popularity Analysis

9. Which are the Top 10 most popular songs?
10. Which are the Bottom 10 least popular songs?
11. What is the average popularity score of all songs?

Release Year Analysis

12. How many songs were released each year?
13. Which release year has the highest number of songs?
14. Which release year has the highest average popularity?

Data Visualization

Create **at least five charts** using Matplotlib.

Mandatory Charts

1. Top 10 Artists by Number of Songs (Bar Chart)
2. Top Genres by Song Count (Bar Chart)
3. Genre Distribution (Pie Chart)
4. Popularity Distribution (Histogram)
5. Songs Released by Year (Line Chart)

Optional Chart

- Top 10 Most Popular Songs (Horizontal Bar Chart)

Business Insights

Write **at least 5 business insights** based on your analysis.

Examples:

- Which artists dominate Spotify?
- Which genres attract more listeners?
- Are recently released songs more popular?
- Which albums have consistently high popularity?
- Which genres require more promotional focus?

Business Recommendations

Provide **at least 3 recommendations** for Spotify.

Example recommendations:

- Promote high-performing genres through curated playlists.
- Collaborate more with artists having consistently high popularity.

- Feature trending albums on the homepage.
- Recommend recently released songs to improve user engagement.

Submission Requirements

Students must submit:

- Python Notebook (.ipynb)
- Cleaned Dataset (if modified)
- Report
- **3 Business Recommendations**
- 5-minute PowerPoint Presentation

Evaluation Criteria

Criteria	Marks
Data Loading & Exploration	15
Data Cleaning	15
Data Analysis	30
Visualizations	20
Business Insights	10
Recommendations	10
Total	100